

Nickel strategic planning outlook to 2050

June 2022



Executive Summary

Long-term forecast

Price scenario

Demand

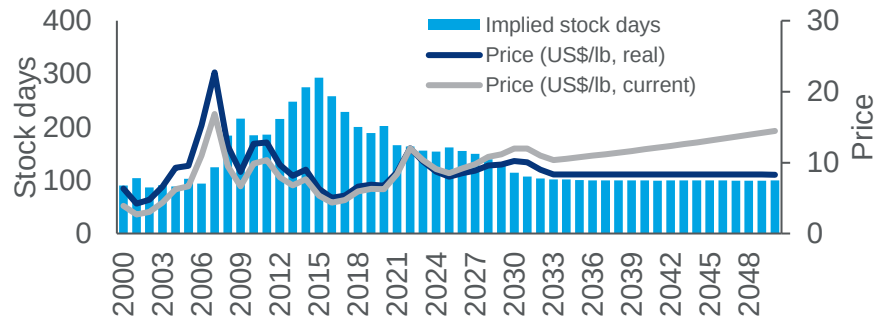
Supply



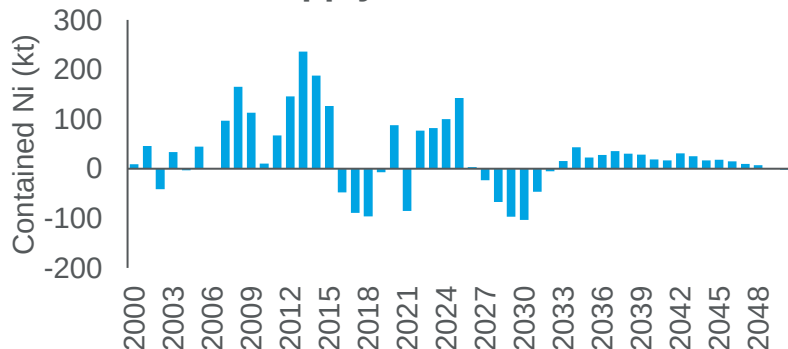
Executive summary – the outlook for nickel

Indonesian expansion drives near-term oversupply before increasing demand from battery segment creates a shortage in late 2020s.

Implied cumulative stock days and nickel price



Global nickel supply-demand balance



- The demand recovery in much of the world will unwind in 2022.
- China will continue to drive global demand through stainless steel and batteries, although primary nickel uptake in stainless will slow as scrap use increases.
- The main risk of a greater surplus comes from ongoing expansion in Indonesian nickel materials production – NPI, MHP and matte.
- The conversion of some NPI lines to matte will affect the dynamics of Chinese/Indonesian nickel supply.
- Mid-term global oversupply will be followed by mounting deficits in late 2020s as strong battery demand continues apace.
- New supply is needed from 2027 increasing to 90-190 ktpa between 2029 and 2036.
- As Chinese companies have already achieved a similar a rate of expansion in Indonesia - and it is principally China that will need all this new nickel - the global target is feasible.
- BUT there will be severe headwinds from the increasing stringency of ESG and the energy transition.
- Nickel prices will fall back from current levels before increasing again to a year average of US\$22,600/t (US\$10.25/lb, real) in 2030, before settling at a long-term price of around US\$18,350/t (US\$8.30/lb, real).



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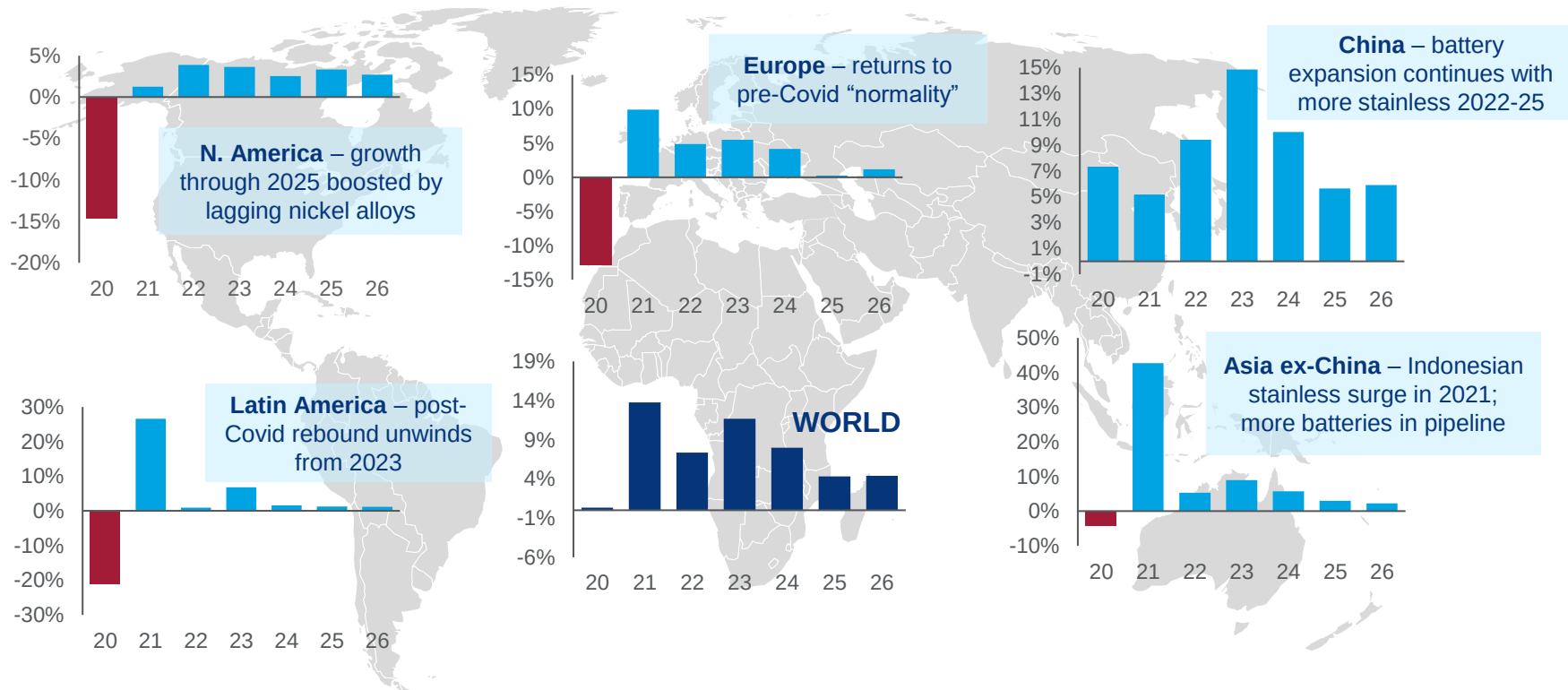
Supply



Global nickel consumption – near term

Robust global growth for the next few years.

Chinese nickel demand remains strong through 2026 as stainless and batteries expand.

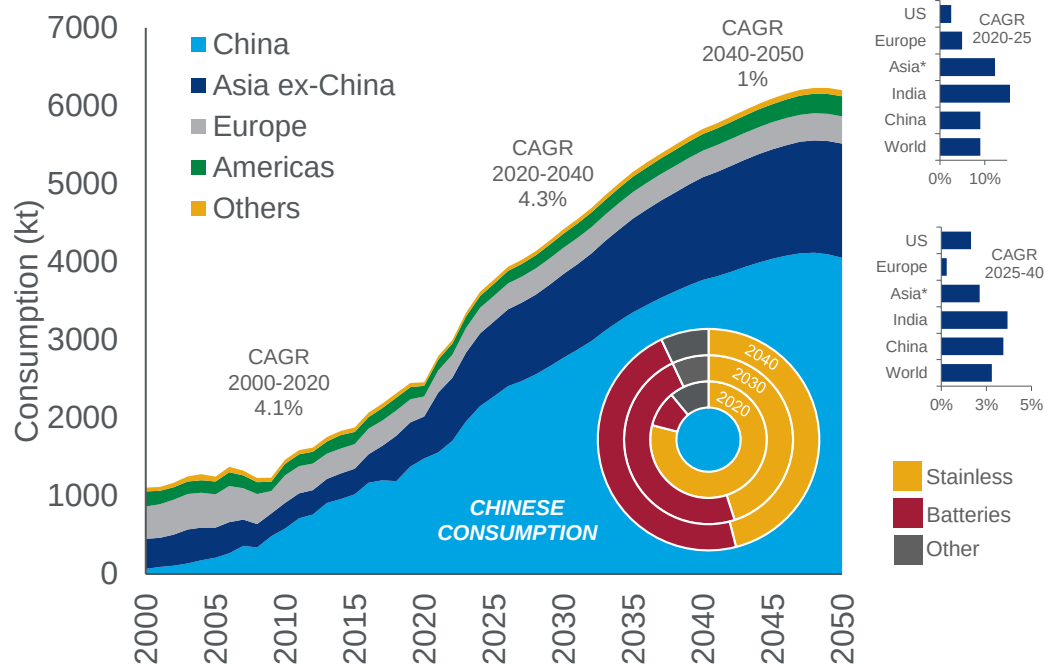




Global nickel consumption – long term

Long-term growth driven by China (stainless and EV batteries) supported by the rest of Asia.

Regional nickel consumption



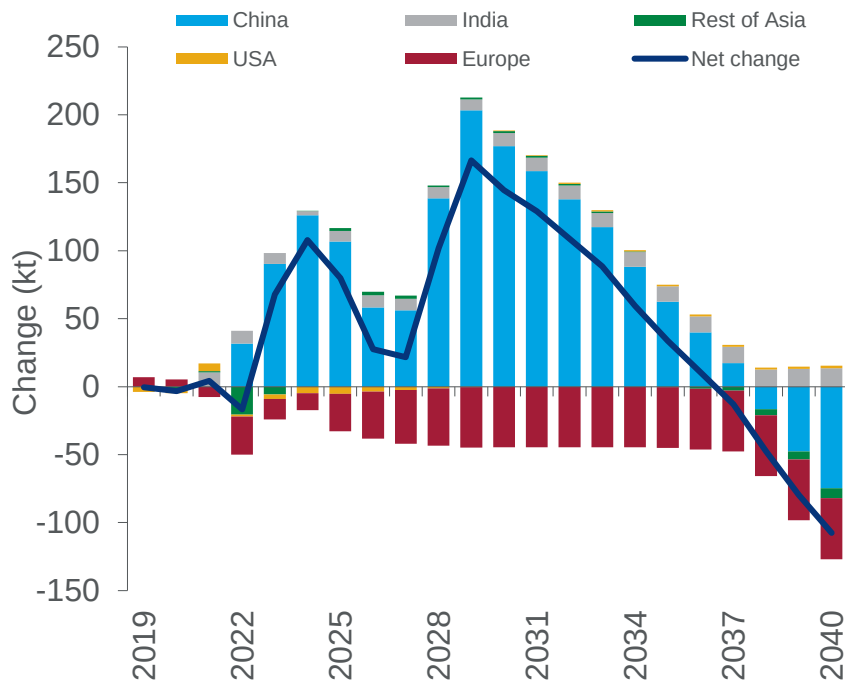
- Global nickel demand up 7% in 2022, to 3.0 Mt, due to sustained demand strength in China and Asia.
- Future growth in stainless:
 - China continues to expand melting capacity.
 - Indonesia ramping up new capacity; more could follow especially if China cancels import duties on Indonesian stainless slab and HRC.
 - Europe protected by duties against Asian imports for five years, and then may gain from a carbon border adjustment.
- Future growth in battery precursor demand:
 - China dominant – battery demand there forecast to triple by 2039, to >1 Mtpa.
 - Further support from South Korean developments. Europe to become a bigger player, facilitated by Chinese/Korean investment.
- China presently accounts for around 55% of global nickel demand due to its dominance stainless and battery precursors. This will increase to around 65% by 2040, due to expansion in the battery segment, and maintained at that level to 2050.



Changes to future nickel demand since the previous quarter

Main changes dominated by upward revisions to Chinese battery precursor demand, supported by near-term expansion in Chinese stainless steel.

Changes to forecast demand 2019-2040



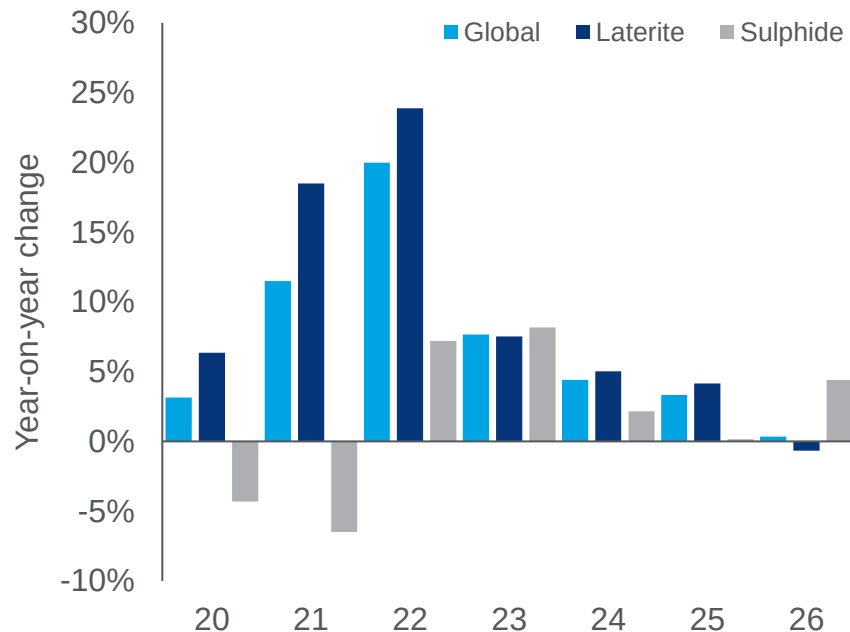
- A revised outlook for China's near-term expansion in stainless steel contributed to the changes in 2022-2025.
- Increases in Chinese mid- to long-term demand are mostly due to substantial upward revision in demand for battery chemicals for the 2028-2037 period.
- This is due to a substantial increase in Wood Mackenzie's forecast sales of BEV and PHEV during that period.
- Reductions in Europe are tied to the impact of the war in Ukraine (near term) and the reduced Ni-in-pCAM demand following the removal of two Johnson Matthey plants from our base case.



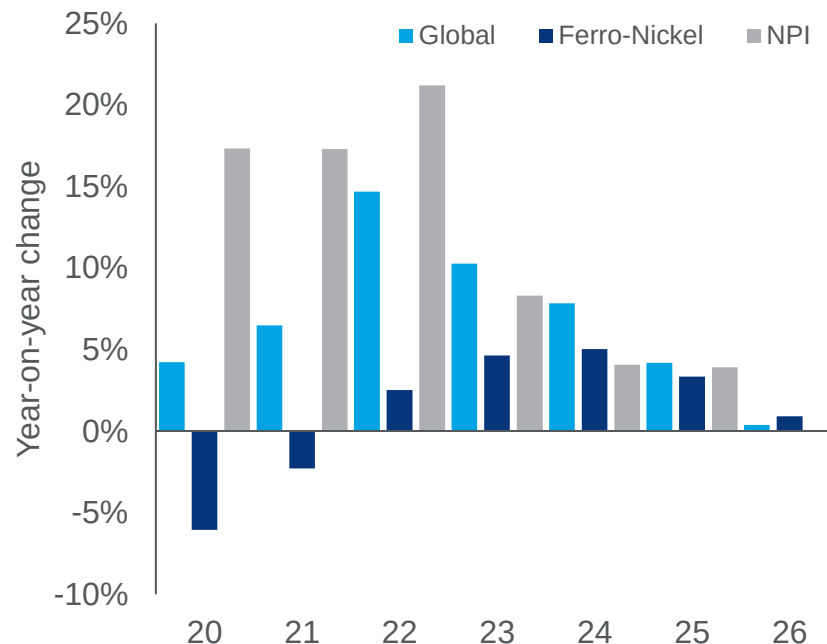
Global nickel production – near term

NPI is still driving near-term production expansion at both the mine and smelter/refinery level. Global mine production to expand by 20.0% in 2022 – primarily due to increasing ore requirement for Indonesia's domestic smelters. New sulphide mines will also provide strong growth in 2022 and 2023.

Global mine production



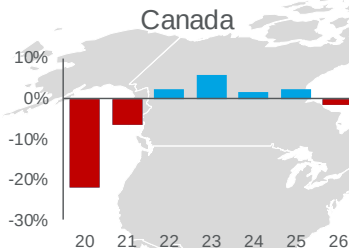
Global refined production



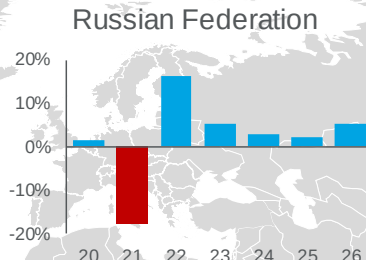


Global mined nickel production – near term

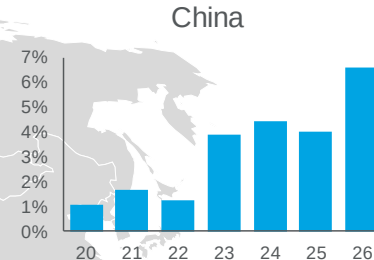
Indonesia is the primary growth area in mined output to supply domestic processing facilities. Australian output is bolstered by new mine projects.



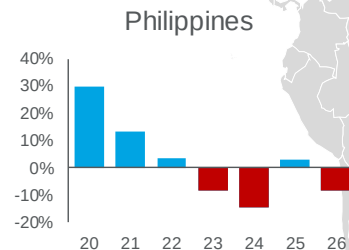
Vale strike action reduced output in 2020. Limited base case growth in mine output. Numerous possible projects could raise output in the long term.



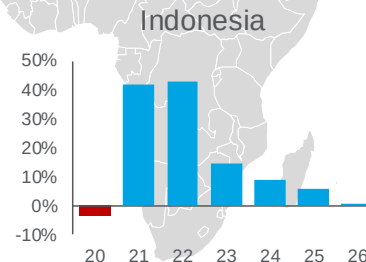
Output in 2021 impacted by flooding in Nornickel mine.



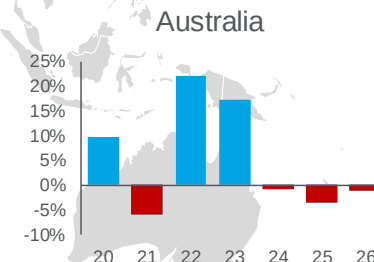
Main growth in China is due to Xiarihamu mine, adding 30 ktpa by 2027.



A sharp fall in Filipino mine output is expected due to reserve depletion. But output has surprised on upside before, and could do so again.



Output down in 2020 due to nickel ore export ban. Growth returned on need to feed domestic smelters.



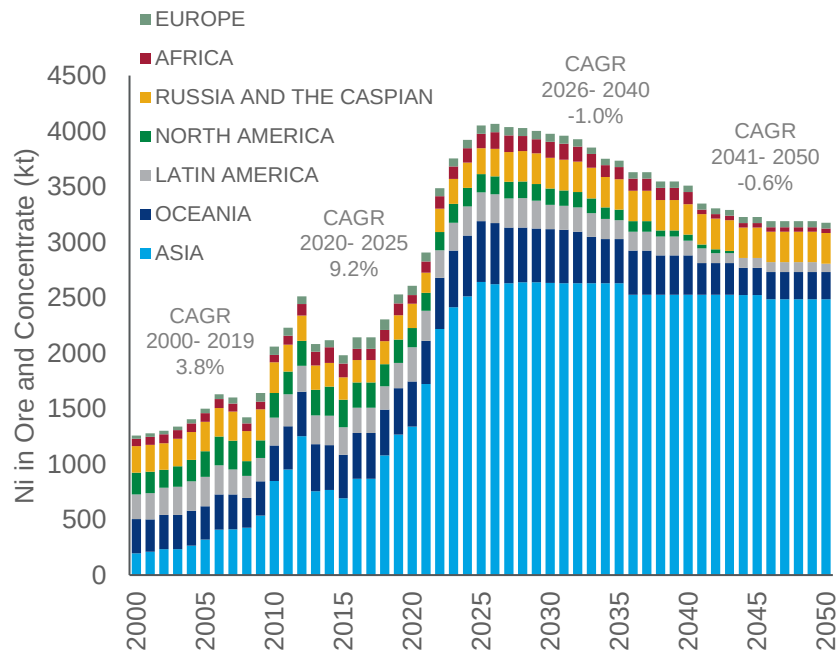
Growth due to Ravensthorpe re-start and, from 2022, Savannah re-start, Odysseus and KNO.



Global mined nickel production – long-term

Indonesia underpins global growth in mine production to feed ongoing NPI, matte and HPAL expansions, with 2022 marking the second year of over 40% annual growth.

Regional mined nickel production 2000-2050

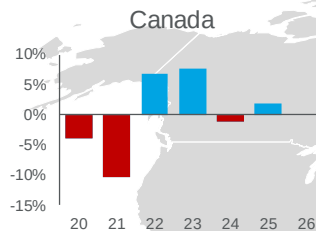


- Global mined nickel production up 20% in 2022, to 3.48 Mt (before adjustments).
- Mid-term growth raises the total to 4.05 Mt in 2025, declining to 3.50 Mt by 2040 and 3.17 Mt by 2050.
- Long-term decline from 2026 through to 2050 is due to reserve depletion at a number of operations.
- Near-term growth continues to stem from Indonesia, with production expanding by over 40% a year in 2022 and 2023. This is to support demand for feed from domestic NPI and matte smelters.
- Forecast Indonesian mine production will reach 2.1 Mt in 2027, and plateau at this level until 2035, before falling to 2.0 Mtpa in the years to 2050.
- The Enterprise mine in Zambia (45 ktpa) is now base case. Where the concentrate will be processed is not yet known.

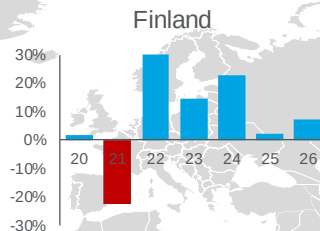


Global refined nickel production – near term

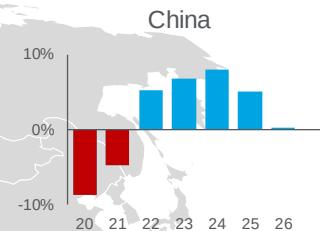
Indonesian NPI continues to bolster near-term growth in finished nickel production.
Nickel matte and MHP from HPAL's will support further growth.



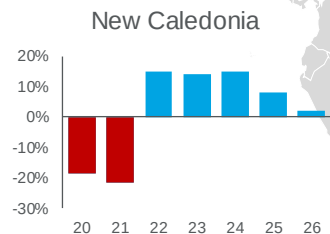
Vale's production returns to "normal" in 2022 after a lack of feed in 2021.



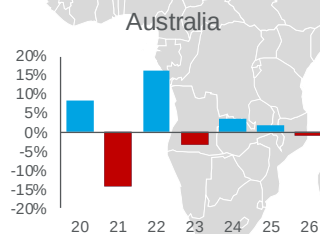
Nornickel is expanding Harjavalta. Terrafame ramping up sulphate production in 2022.



Chinese refined nickel output to increase from 2022 as Quzhou Huayou expands chemical production.



Greater output at Koniambo and Doniambo expected from 2022.



Double-digit increase in refined output as Kwinana expands nickel sulphate production in 2022.



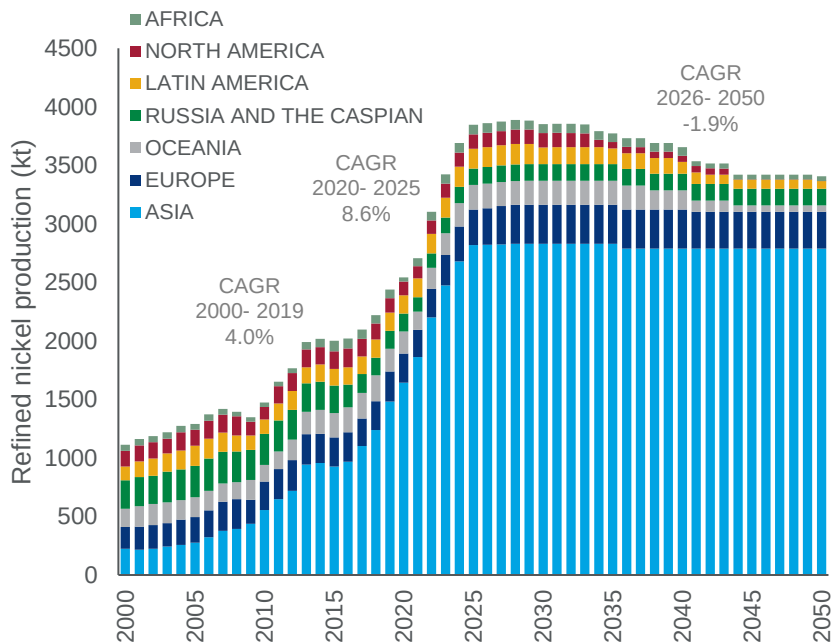
Refined output to exceed 1.2 Mt in 2022 as VDNi and Weda Bay expand NPI (220 kt) and PT Gunbuster starts up.



Global refined nickel production – long term

NPI production in Indonesia and chemical production in China dominates refined supply.

Regional refined nickel production 2000-2050



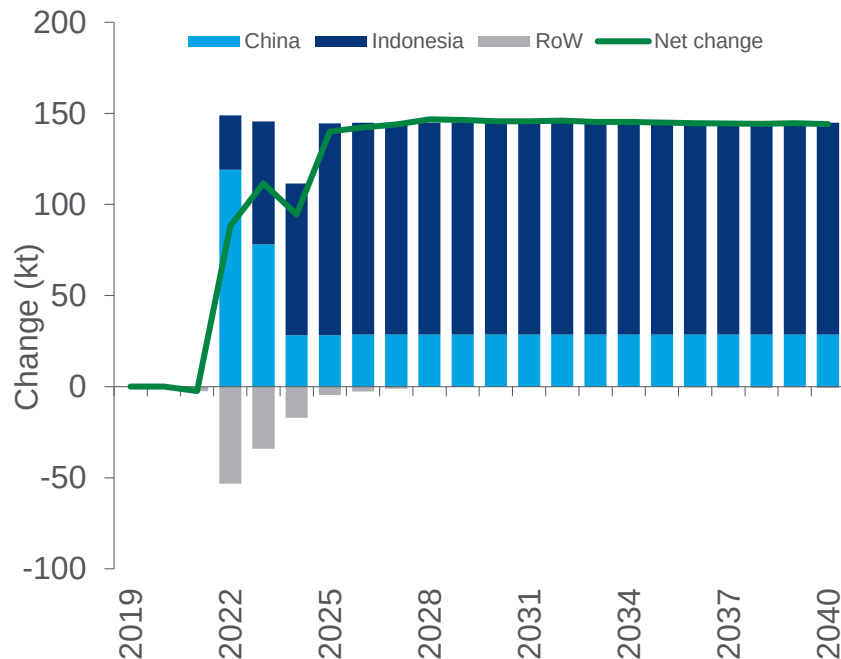
- Global refined nickel to increase by 14.7% in 2022 to 3.10 Mt. Growth continues strongly to 3.85 Mt in 2025.
- After 2025, refined production will plateau at this level before a gradual decline to 3.41 Mt in 2050, due to reserve depletion.
- Secondary nickel supply will come from recycling of battery manufacturing scrap and end-of-life batteries. This will provide a significant boost to future nickel production.
- Forecast secondary nickel supply increases to 130 kt by 2030, rising to 686 kt by 2040 and 1.27 Mt by 2050.



Changes to future refined nickel production since the previous quarter

Main revisions since Q1 relate to Indonesian NPI, Chinese chemicals output and an extension of the mine life at Koniambo.

Changes to forecast production 2019-2040



➤ Main changes are:

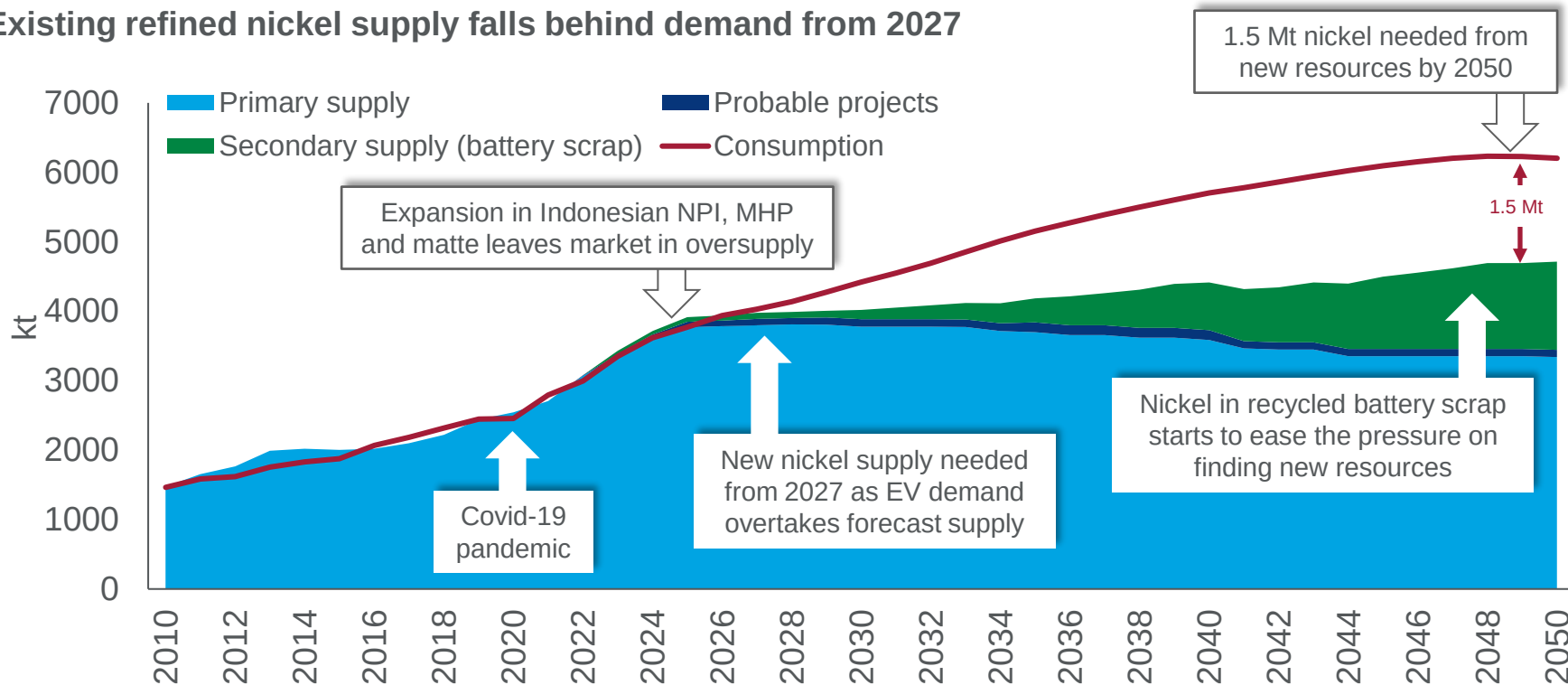
- Araguaia FeNi moved to base case.
- Ferronikeli taken off-line due to high energy costs. New owner Yildirim Group will be keen to re-start production.
- Weda Bay – forecast production revised up steeply - +120 ktpa.
- VDNi also revised up while Tsingshan IMIP revised down.
- Chinese NPI revised up as production rates so far this year are higher than expected. This is supported by continued strength of ore shipments from the Philippines.



The long-term shortfall in supply

If China/Indonesia supplies all of China's ongoing battery needs, the burden of future supply falling the RoW will need a long-term price of around US\$18,350/t (US\$8.30/lb, real).

Existing refined nickel supply falls behind demand from 2027





Risks to the nickel price outlook

Prices bullish on electrification outlook but Indonesian NPI leaves market in surplus.

Short term

Long term

Upside factors

Downside factors

- Chinese nickel demand in batteries and stainless is stronger than forecast, offsetting wider macro-economic concerns and countering projected surpluses.
- Supply tightness of goods and services from renewed Covid outbreaks (in China) and war in Ukraine (Europe) accentuates demand.

- Government and company targets for electrification imply a larger global EV fleet coming sooner than forecast, which will mean greater high-nickel battery requirements.
- ESG concerns around emissions and waste disposal slow nickel developments in Indonesia, thereby tightening supply.
- Ongoing reluctance of western producers to invest in nickel projects to meet battery demand.

- Widespread economic and inflationary concerns undermines demand; consumers become more choosy as to what to spend their money on.
- The ongoing ramp up in Indonesian NPI, MHP and matte pushes the market further into oversupply and ensures future availability of nickel for batteries.
- Further Covid outbreaks restrain the Chinese economy and undermine sales of EVs.

- Collection and distribution of stainless steel scrap increases in China as more industrial “old” scrap becomes available. Chinese scrap ratios go higher than forecast thereby reducing projected demand for primary nickel.
- New battery chemistries that depend on less nickel or none at all, become competitive.



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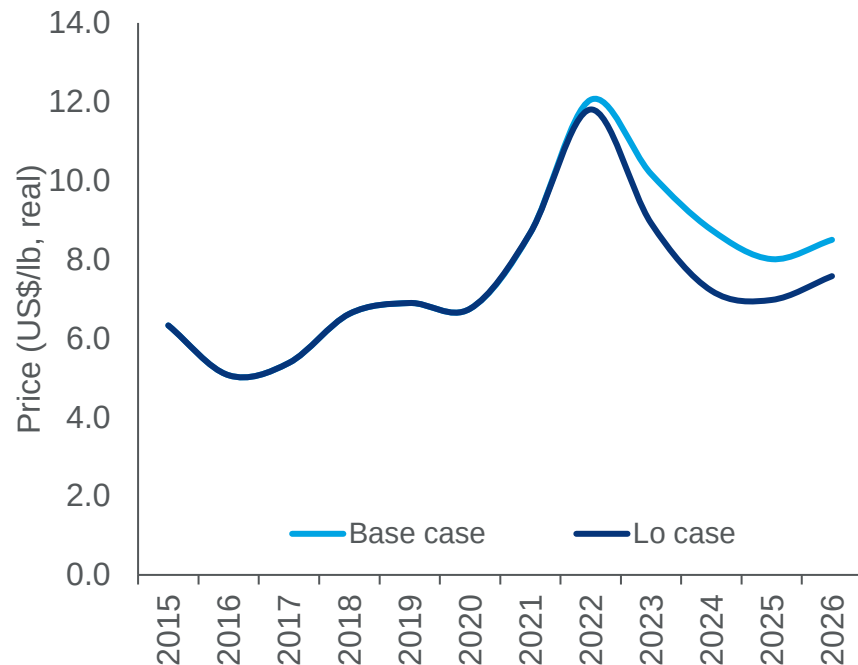
Demand
Supply



Price scenarios

Lo case – global economic concerns, especially regarding China and the war in Ukraine, turn sentiment bearish and LME prices return to pre-“Big Short” levels.

Scenario implications for nickel prices



- Our Base Case assumes that nickel prices will fall as the post-pandemic demand recovery unwinds and the continuing rapid expansion in Indonesian nickel production pushes the market into oversupply.
- Our Lo Case assumes that nickel prices will fall sooner and steeper than the base case, due to:
 - weaker demand caused by accelerating economic concerns tied to Covid risk in China and the war in Ukraine.
 - the return of nickel prices to January 2022 levels (ie, pre-LME trading halt) by the end of the year.

	2018	2019	2020	2021	2022	2023	2024	2025	2026
Base case									
Stock days	201	189	202	166	164	156	154	162	155
Nominal US\$/lb	595	631	625	838	1205	1038	911	850	920
Real US\$/lb	662	690	675	867	1205	1017	875	801	850
Lo case									
Stock days	201	189	202	166	171	171	168	174	167
Nominal US\$/lb	595	631	625	838	1180	910	750	740	820
Real US\$/lb	662	690	675	867	1180	892	721	697	758



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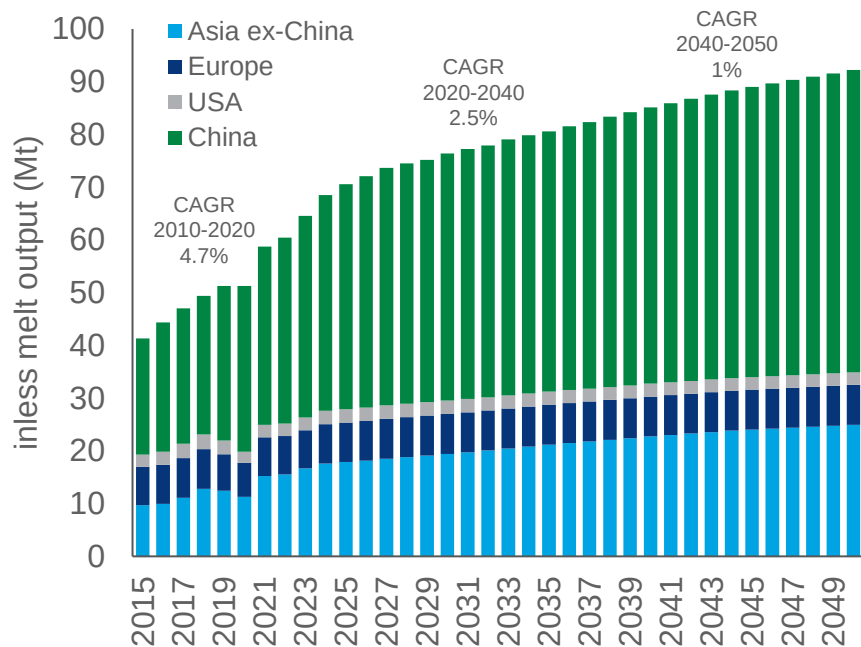
Supply



Global stainless – robust growth as China keeps adding capacity

Global CAGR to moderate as growth in China slows down.

Global stainless melt production by region



The outlook for regional stainless

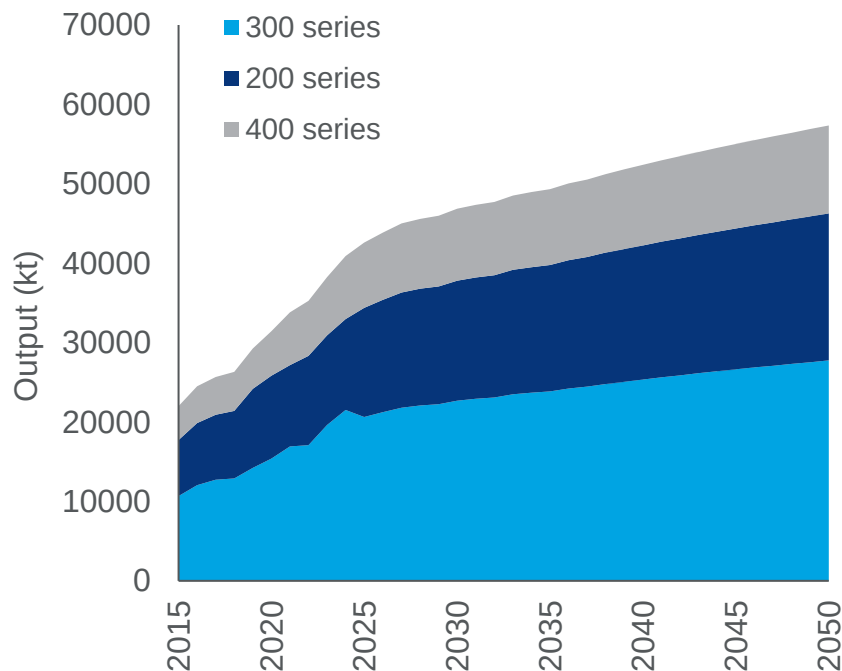
	Long-term growth in stainless production dominated by China and Indonesia
	China 1.8 Mt new melt shop capacity just started plus further 2.5 Mtpa possible before year end. More than 3 Mtpa in the pipeline by 2025. Output up 4% to 35.2 Mt in 2022, increasing to 52 Mt in 2040 and 57 Mt in 2050. CAGR 1.7%. Scrap use to rise, partly determined by availability of Indonesian NPI.
	Indonesia Tsingshan producing at 4 Mtpa, Delong Obsidian halfway to 3.5 Mtpa. New projects likely if Chinese duties against Tsingshan's semis are eased.
	India up 3.4% in 2022, to 4.1 Mt, before heading to >10 Mtpa over the long term. CAGR 3.5% to 2050.
	Other Asia – production broadly flat over the long term at approx. 6.2 Mtpa
	Taiwan: Domestic production partly replaced by Indonesian imports. Future investment in Indonesia possible. South Korea: Posco at full capacity. Outlook steady at 2.4 Mtpa. Japan: Production steady at around 3 Mtpa.
	Europe – Small contraction in 2022 to 7.2 Mt before recovering to 7.5 Mtpa over long term. Duties against Asian imports in place for five years. Carbon border adjustment may boost protection later.
	USA – Post-Covid rebound supports growth from 2.4 Mt in 2021 to 2.8 Mt by 2025 (CAGR 4%). Slight long-term decline thereafter.



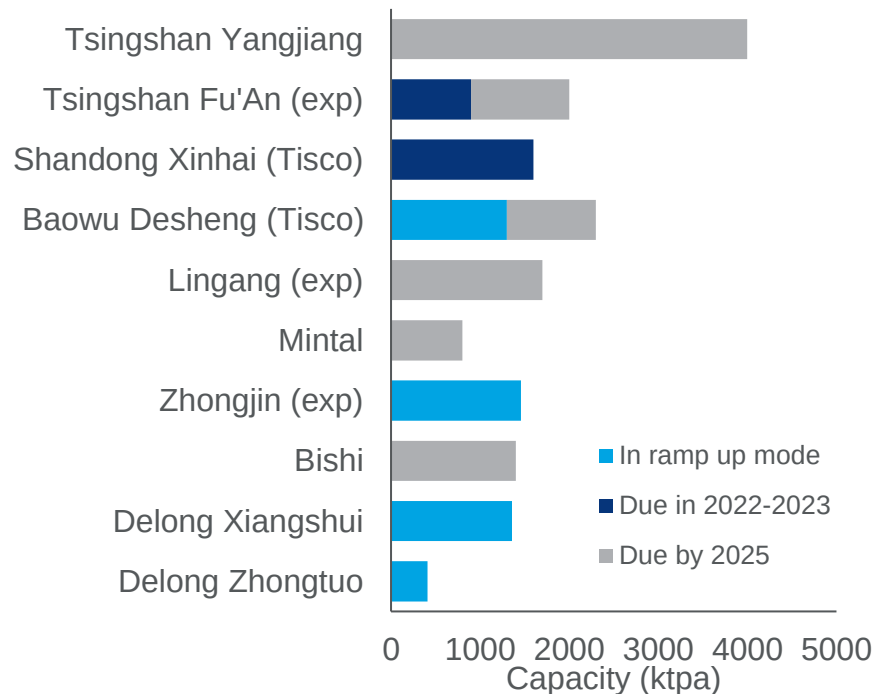
China stainless – further expansion over the next five years

High-Ni 300 series to remain dominant. Scrap use to increase. Race for market share to drive greater interest in specialties, including non-Ni 400 series; austenitic ratios could fall.

China stainless production outlook, by grade



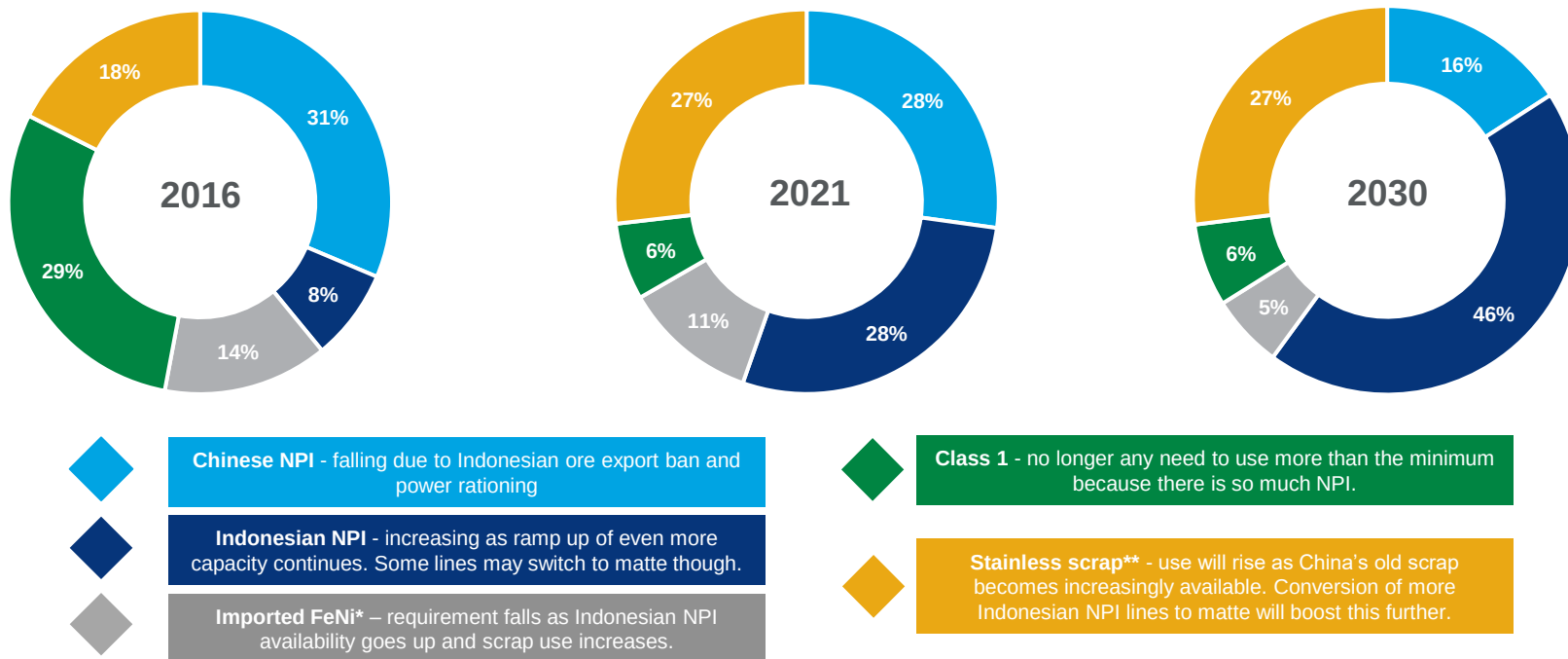
Melting capacity in the pipeline





China stainless – becoming self-sufficient in nickel

Indonesian NPI has reduced China's dependence on Class 1 nickel. As “old” scrap becomes more available, so FeNi imports will also fall. Any limitation on NPI – such as line conversion to matte for batteries – will only boost more scrap collection.

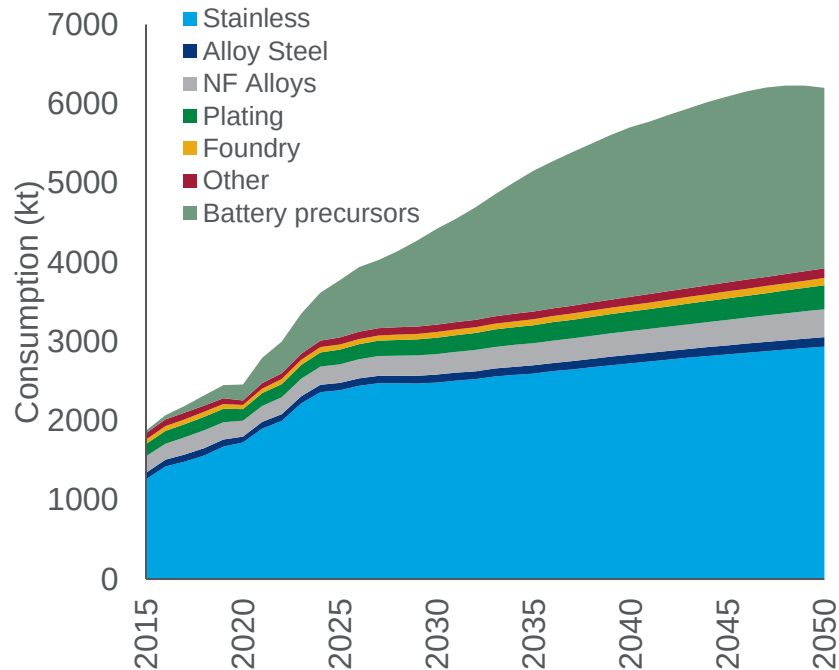




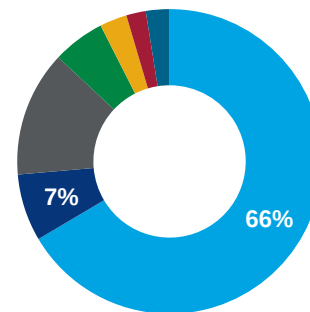
Nickel consumption by first use – the future is batteries!

Stainless remains the main first use of nickel, although its share falls to 47% by 2050, as battery precursors for EVs and energy storage become increasingly important.

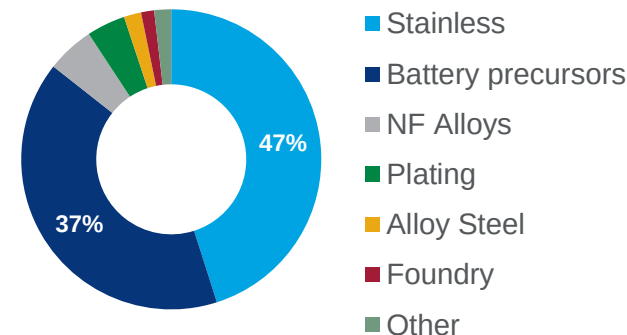
Global nickel demand by first use



2022



2050



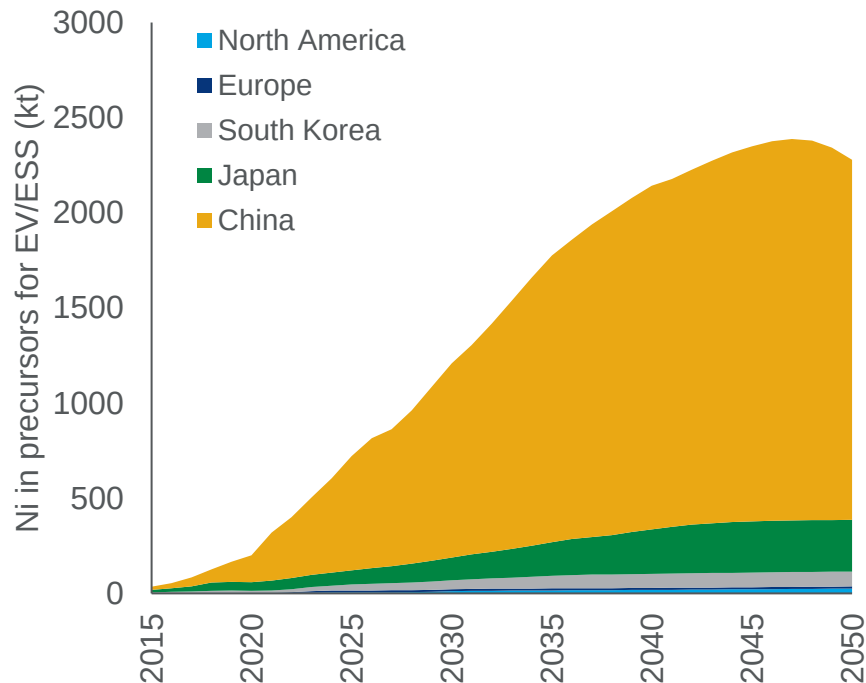


Global demand for nickel sulphate in batteries rising steeply

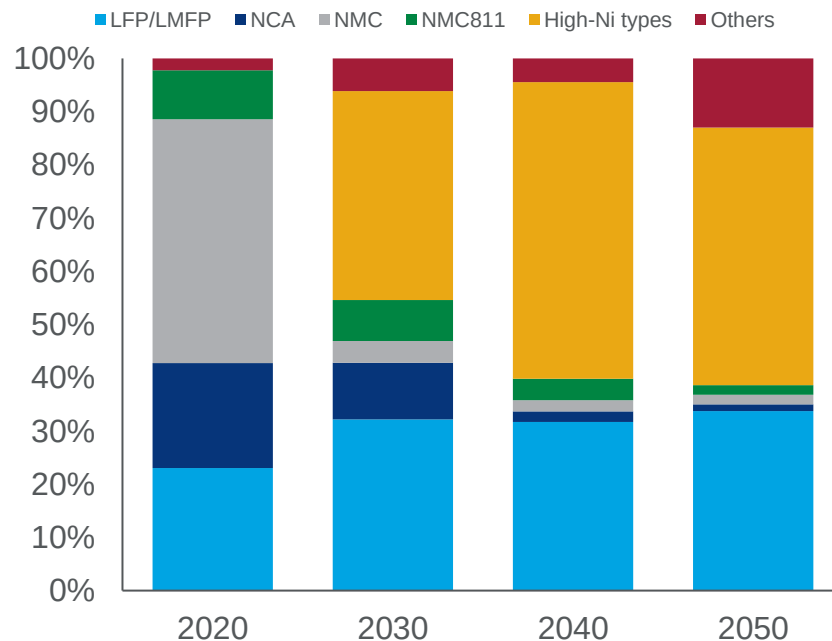
China dominant in both NiSO_4 and PCAMs/CAMs production.

Nickel chemistries will evolve and remain dominant over the long term.

Nickel demand in battery precursors



Future change in battery chemistries

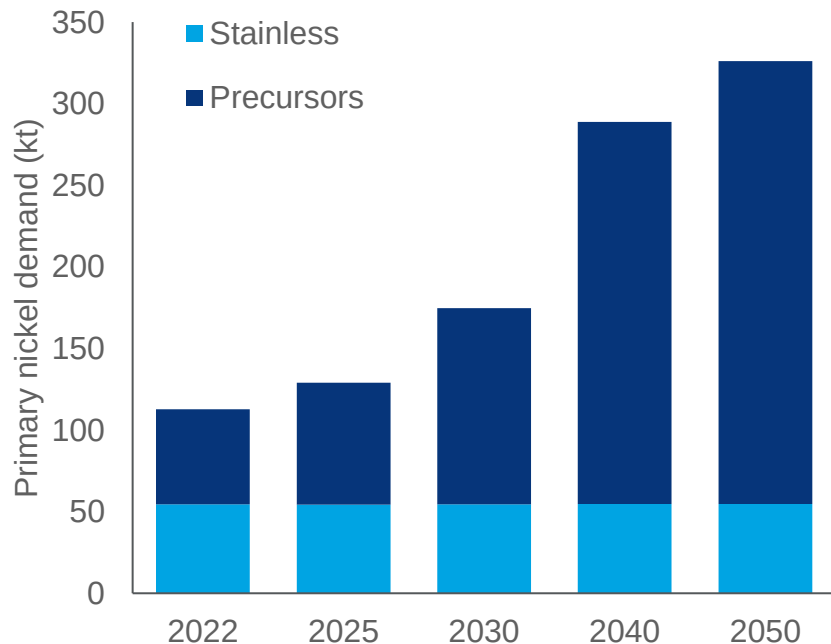




Japan's nickel demand in pCAMs is at risk; Indonesia on the up

Despite leadership in NiMH and Li-ion, Japan has stopped investing in an electric future. In contrast, Indonesia has attracted partners with US\$15 billion for projects in April.

Forecast Japanese nickel demand in pCAMs: Should we continue to expect it?



- Japan's nickel in stainless demand flatlines due to low growth in austenitic demand. Strengths lie in niche duplex and trademarked ferritic grades that compete with high-Ni stainless and alloy steels.
- Our assumption of strong growth in demand from rechargeable batteries is at risk. Despite Japan leading advances made in this area since the 1990s, there is little new investment in raw materials or cell capacity today. There is no appetite to take on the technical challenges of recycling.
- This lack of EV/ESS investment in Japan could be countered by South Korean international investments and developments in Indonesia.
- Indonesia announced US\$15 billion investment over two projects: Antam and PT Industri Baterai Indonesia will work with CATL and Brunp from China, and LG and Hyundai from Korea, in projects covering mines, sulphates, pCAMs to CAMs, and cells through to electric vehicles. We await further clarification, but combined output could be 70-75 kt nickel in sulphate.



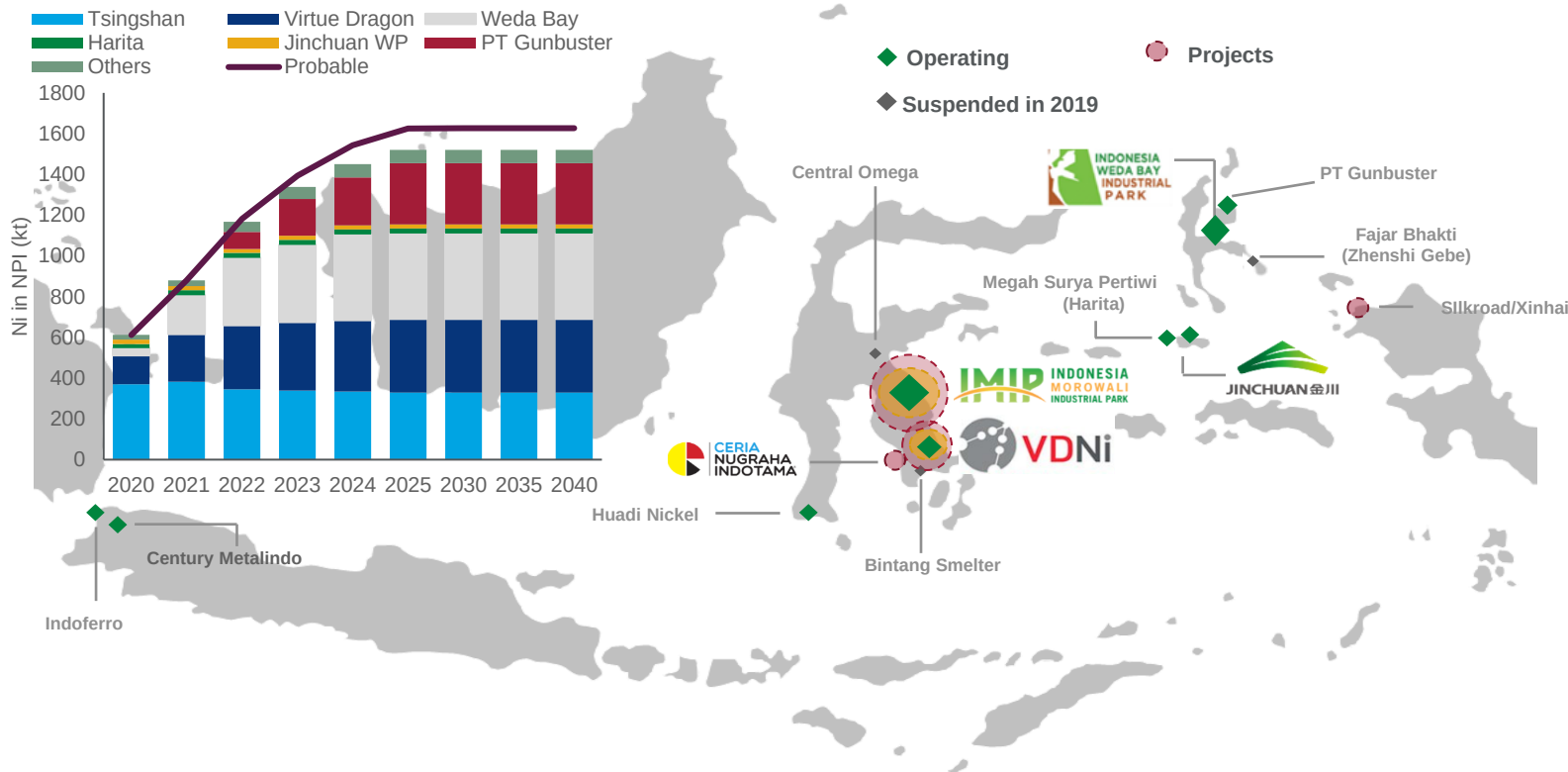
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Indonesian nickel-in-NPI production will reach 1.62 Mt by 2025

The start up of PT Gunbuster will add 300 kt nickel to the market, while Tsingshan IMIP, Virtue Dragon and Weda Bay will account for another 1.1 Mt between them by 2025.

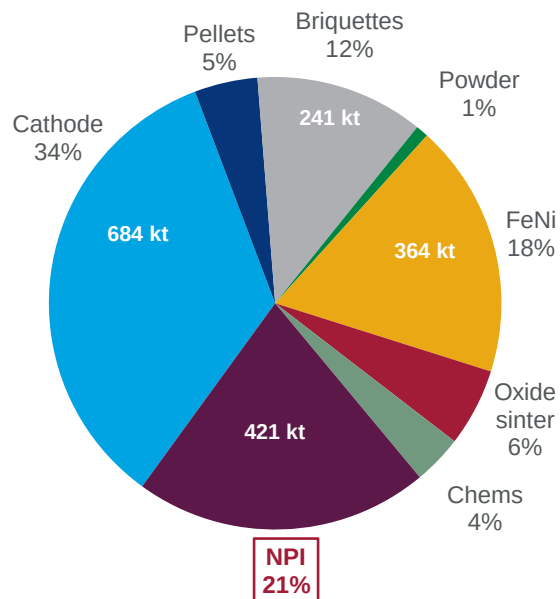


Nickel products by type from 2015 to 2030: NPI continues to dominate

But growth in chemicals will account for 15% of the market by 2030

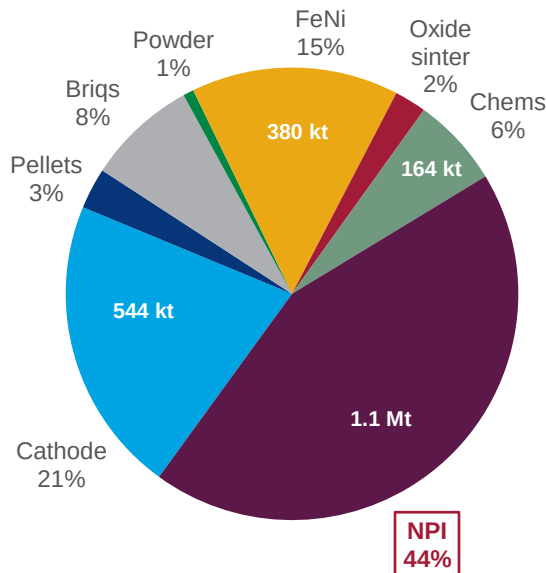
Class 1 nickel held a 52% market share..

2015 = 2.0 Mt



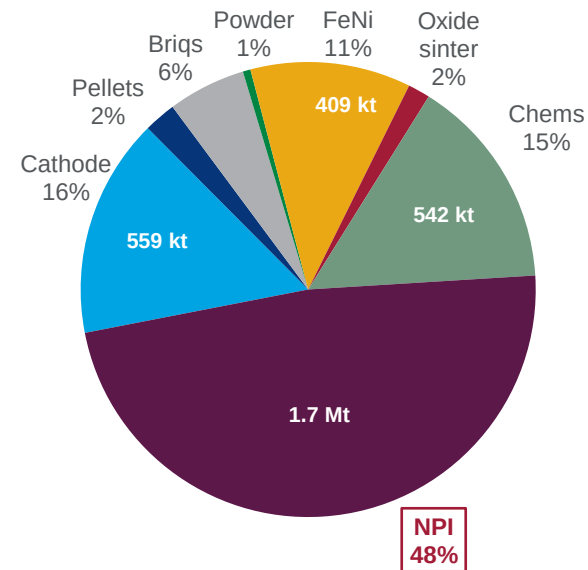
..but by 2020 this had fallen to 32%, as NPI output doubled...

2020 = 2.56 Mt



...and will rise by another 600 kt by 2030 to claim 48% of the market

2030 = 3.58 Mt

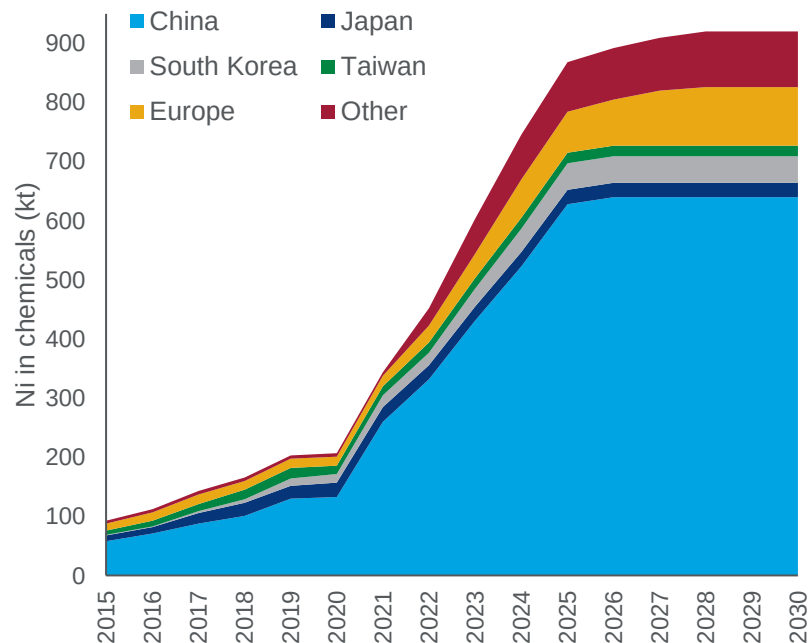




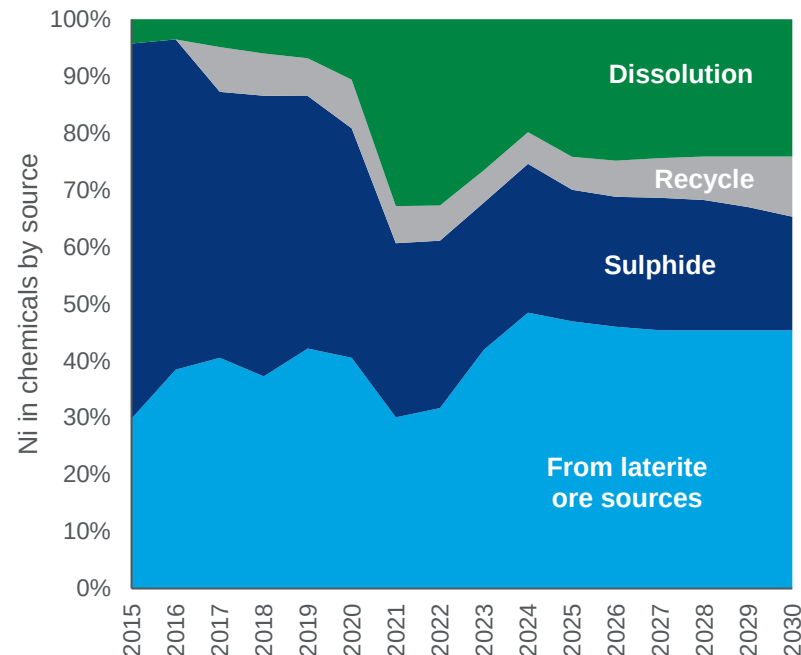
Nickel in chemicals production

China's nickel sulphate production to climb exponentially in the years to 2025, while European growth will be helped by Chinese and South Korean investments

Global nickel sulphate production, 2015 - 2030



Nickel sulphate feed source, 2015 - 2030





Probable mine projects

Enterprise is the first sizeable sulphide project to come on-line in years despite “high” nickel prices and a positive outlook, which in turn have been propelled by the energy transition and burgeoning demand for EVs.

There is still seemingly limited appetite for investment outside Indonesia.

Project	Country	Status	Ore Type	2022	2023	2024	2025	2026	2027	2028	2029	2030	2035	2040	2050
Black Swan/Silver Swan/Golden Swan	Australia	Probable	Sulphide	2	8	8	8								
Wulantaolegai	China	Probable	Sulphide		0	2	3	3	3	3	3	3	3		
Moa Bay	Cuba	Probable	Laterite				2	5	5	5	5	5	47	47	
Probable Total				2	8	10	13	8	8	8	8	8	50	47	0

Source Wood Mackenzie and company reports

- Sherritt is considering an expansion of Moa Bay. More details will be announced later this year so we await criteria on funding before moving to base case.
- Horizonte announced a US\$633 million funding package and started building the Araguaia FeNi project.
- First Quantum started developments to bring its Enterprise nickel project to production. As yet it is unclear where the concentrate will be processed.



Probable refinery projects

Indonesia's dominance continues with additional NPI anticipated from Tsingshan at Morowali (IMIP) from 2023. Yin Hai Wanxiang and Phase 2 projects at PT Huadi and PT Halmahera will also provide more nickel from NPI and MHP, respectively.

Project	Country	Status	Product	Class	2022	2023	2024	2025	2026	2027	2028	2029	2030	2035	2040	2050
Fort Saskatchewan	Canada	Probable	Metal	Class 1				2	5	5	5	5	5	40	40	
Rochester	United States	Probable	Chemicals	Class 2		2	4	4	4	4	4	4	4	4	4	4
IMIP	Indonesia	Probable	NPI	Class 2	6	30	40	40	40	40	40	40	40	40	40	40
PT Huadi Nickel Alloy Indonesia (Phase 2)	Indonesia	Probable	NPI	Class 2			10	20	20	20	20	20	20	20	20	20
Yin Hai Wanxiang	Indonesia	Probable	NPI	Class 2	8	16	32	32	32	32	32	32	32	32	32	32
PT Halmahera Persada Lygend - Ph 2	Indonesia	Probable	MHP			9	12	14	14	14	14	14	14	14	14	14
Probable Total					14	57	97	111	114	114	114	114	114	150	150	110
of which					14	46	82	92	92	92	92	92	92	92	92	92
FeNi + NPI																

Source Wood Mackenzie and company reports

- » While the project pipeline we currently consider as “probable” projects is limited in its potential to supply nickel units to the battery sector, some NPI lines could potentially now be matte lines. Certainly more repurposed NPI lines, as well as purpose built lines to produce matte, are being considered in Indonesia (see next slide). These changes could provide significant tonnages of intermediate material as feed to the battery sector in China.
- » LG Energy Solution has “broken ground” on a US\$3.5 billion, 150 kt nickel sulphate plant in Batang, Indonesia. This will be included in our Base Case outlook for next month. This will feed an integrated precursor and cathode plant to be constructed at a cost of US\$2.4 billion on the same site. Ore will be supplied by Antam (ca. 16 Mwmt a year) from a new mine to be developed in East Halmahera.



Nickel intermediate production in Indonesia

The potential production of nickel intermediates from Indonesian projects to feed the battery sector is huge. Over 1 Mt of potential nickel capacity is now either operating or proposed

Name	Location	Principal partners	Process	Product	Capacity (kt Ni/a)	Capex (US\$M)	Capital intensity (\$/kt)	Possible start date	Status
PT Halmahera Persada Lygend - Ph 1	Obi Island	Ningbo Lygend, Harita	HPAL	MHP/Sulphate	37	1063	28529	2021	Production
PT Halmahera Persada Lygend - Ph 2	Obi Island	Ningbo Lygend, Harita	HPAL	MHP/Sulphate	17	437	25706	2022	Probable
PT Halmahera Persada Lygend - Ph 3	Obi Island	Ningbo Lygend, Harita	HPAL	MHP/Sulphate	60			2023	Possible
PT QMB New Energy	Morowali	GEM	HPAL	Sulphate	50	1600	32000	2022	Base Case
Huayue Nickel & Cobalt	Morowali	Huayou	HPAL	MHP	60	1600	26667	2021	Production
PT Huafei	Weda Bay	Tsingshan, EVE, Huayou	HPAL	MHP	120	2080	17333	2025	Base Case
PT ZhongTsing New Energy Ph 1	Morowali	CNGR, Rigqueza International	Smelting	Matte	30	243	8100	2022	Base Case
PT ZhongTsing New Energy Ph 2	Morowali	CNGR, Rigqueza International	Smelting	Matte	30	330	11000	2022	Base Case
Tsingshan	Morowali	Tsingshan	Smelting	Matte	75			2022	Production
PT Youshan Nickel Indonesia	Weda Bay	Huayou Cobalt, Tsingshan	Smelting	Matte	34			2022	Production
Hengjaya/Ranger*	Morowali	Nickel Industries Limited	Smelting	Matte	20				Probable
Batang	Java	LG Energy, Antam	HPAL	Sulphate	30	3500	116667	2025	Base Case
PT Huake Nickel	Weda Bay	Huayou Cobalt, Tsingshan	Smelting	Matte	45			2023	Probable
PT ChengMach Nickel	Weda Bay	Chengtun, Extension Investment	Smelting	Matte	40	350	8750	2023	Possible
Weiming/Indigo		Weiming/Indigo	Smelting	Matte	40	400	10000	2023	Possible
Pomalaa	Pomalaa	Huayou Cobalt, PT Vale	HPAL	MHP	120			2026	Possible
BASF/Eramet	Weda Bay	BASF/Eramet	HPAL	MHP	42			2030	Possible
PT Ceria Nugraha Indotama	Obi Island	PT Ceria	HPAL	MHP	40			2030	Possible
Jinchuan	Obi Island	Jinchuan	HPAL	MHP	20			2030	Possible
PT Antam	East Halmahera	PT Antam	HPAL	Sulphate	75	6400	85333	2030	Possible
Total MHP/Sulphate potential capacity					671				
Total Matte potential capacity					314				

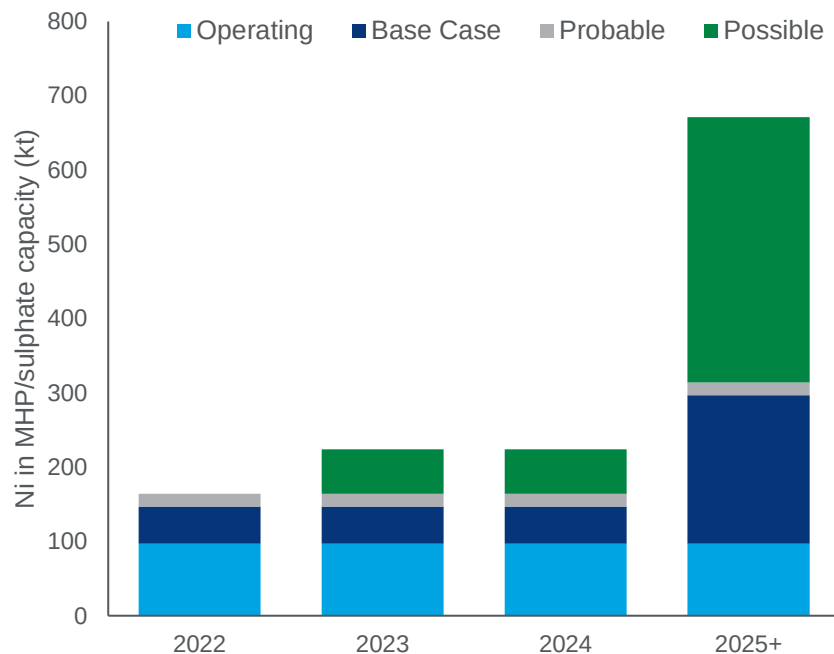
* Most likely forms part of the Tsingshan 75kt announcement rather than additional



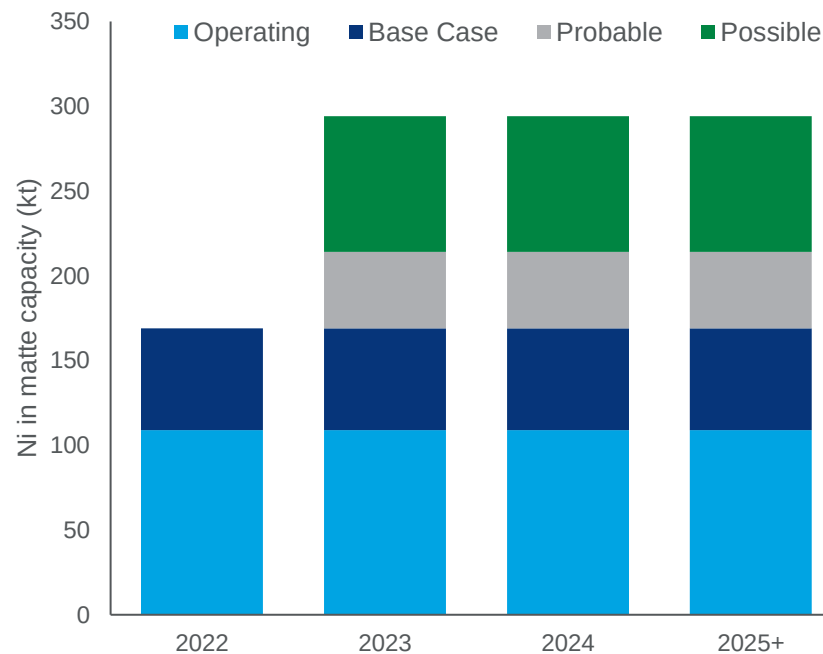
Nickel intermediate production in Indonesia

Potential for over 1 Mt of nickel in intermediates production being brought on-line in the next five years

Nickel in MHP/sulphate capacity



Nickel in matte capacity





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